

UniTS Anomaly Ratio Sweep

ESA-Mission2 | Generated 2026-03-12 10:53

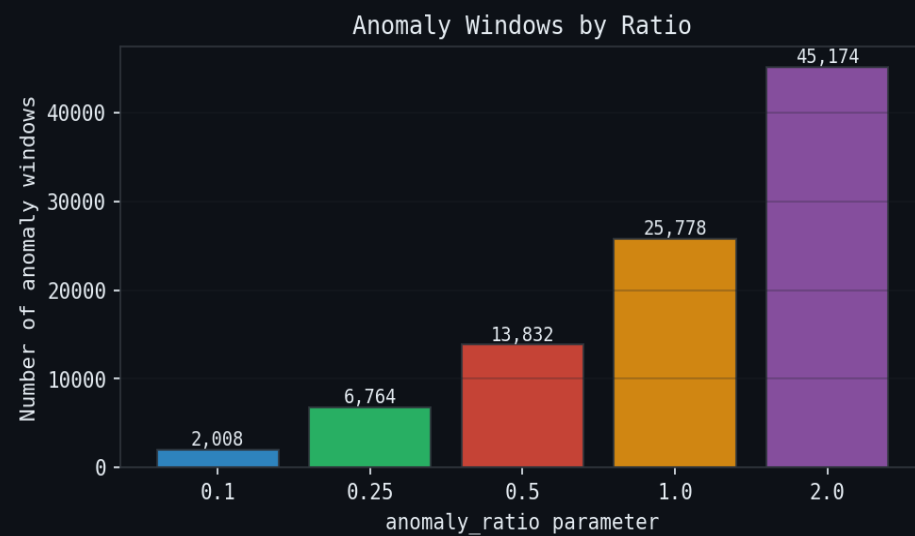
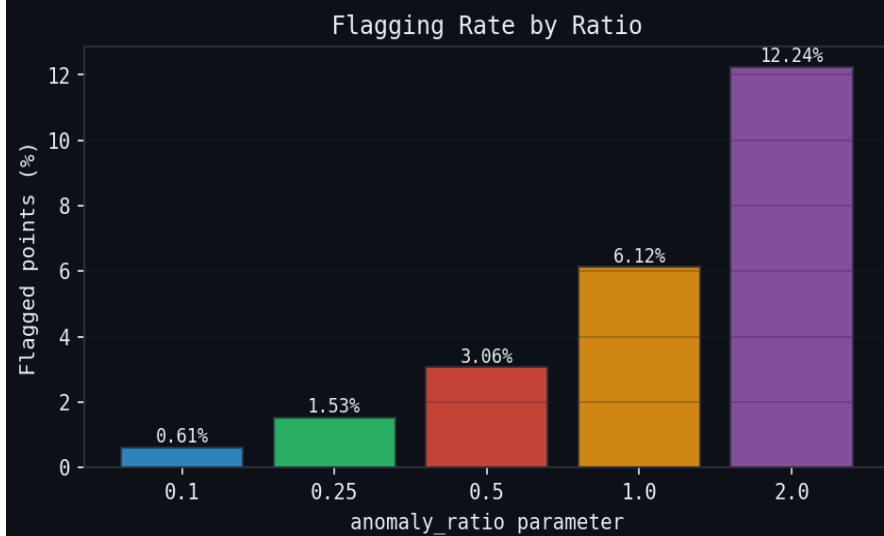
Results by Ratio

anomaly_ratio	Threshold	Flagged Points	Flag Rate	Windows	Precision	Recall	F1
0.1	0.235167	4,813	0.612%	2,008	0.9272	0.9452	0.9361
0.25	0.142498	12,034	1.531%	6,764	0.7812	0.9796	0.8692
0.5	0.101341	24,068	3.061%	13,832	0.6139	0.9869	0.7570
1.0	0.070704	48,136	6.122%	25,778	0.4290	0.9917	0.5989
2.0	0.045301	96,272	12.245%	45,174	0.2679	0.9990	0.4225

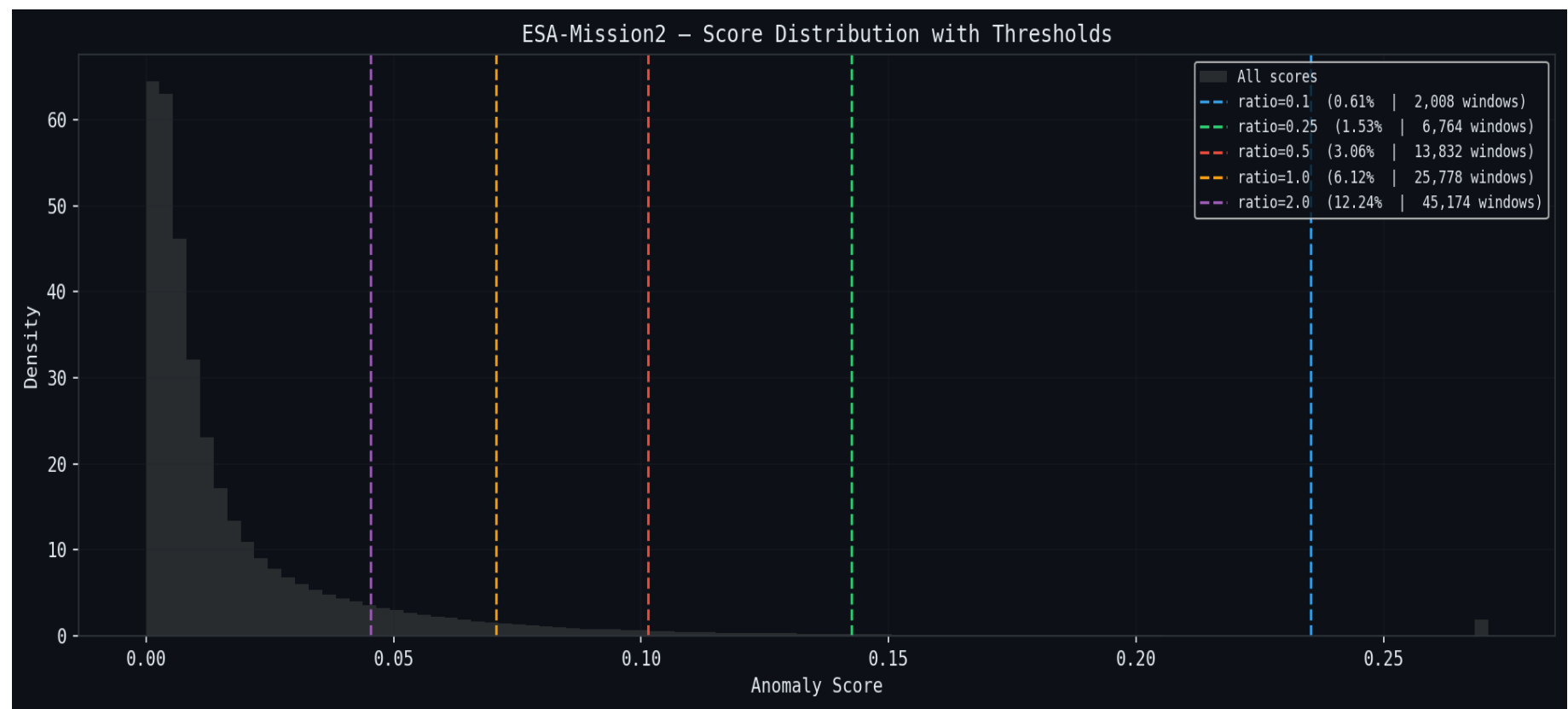
anomaly_ratio controls the detection threshold: UniTS flags the top ratio% of reconstruction error scores as anomalies. Lower values = stricter threshold = fewer flags. The threshold column shows the exact score cutoff for each ratio.

Flagging Rate & Window Count

ESA-Mission2 – Ratio Sweep Summary



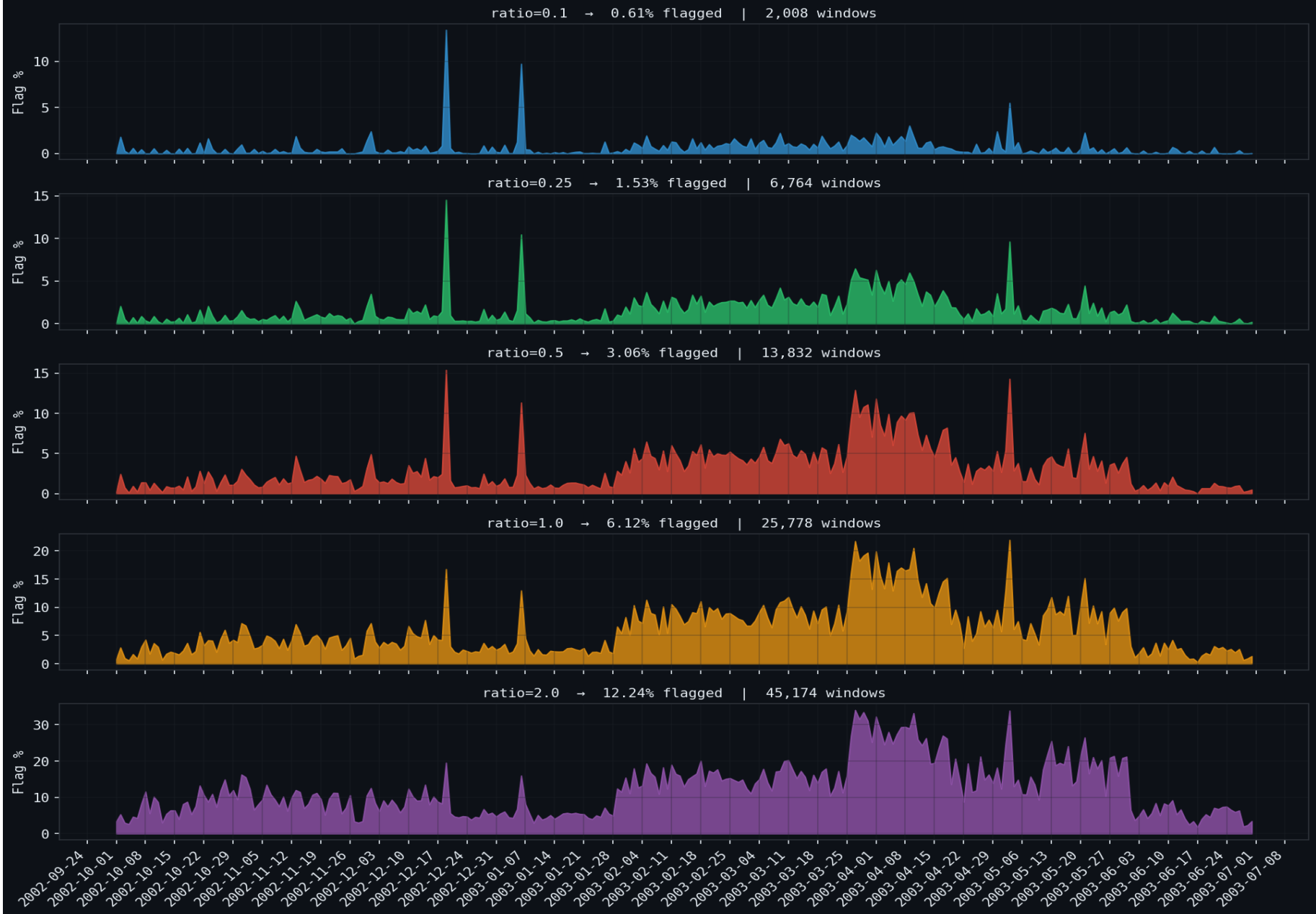
Score Distribution with Thresholds



Dashed vertical lines show where each ratio sets its threshold. Choose a ratio whose threshold sits clearly in the tail of the distribution rather than in the main body of normal scores.

Daily Flagging Rate by Ratio

ESA-Mission2 – Daily Flagging Rate by Ratio



A realistic anomaly rate for a nominal satellite is typically 0.1–1%. Look for a ratio that shows isolated spikes rather than sustained elevated flagging.